Week 6 Deliverable: Functional Requirements & Acceptance Criteria

FR ID	Module / Area	BR ID	Functional Requirement Specification	Acceptance Criteria	Priority
FR-01	Intake Interface	REQ-01	**Mandatory Field Flagging & Submission Blocking**: The UI needs to visually flag all mandatory fields with an asterisk (*). If any mandatory field is empty or unselected, the system blocks the submission and highlights it red.	• All buttons are grayed out until all mandatory fields are filled out except if red-flag condition has been triggered	**Must-Have**
FR-02	Intake Interface	REQ-01	**Standardized Dropdowns & Controlled Lists:** The UI restricts data entry to standardized dropdown lists for: • Country of Residence (UN countries) • Tax Domicile (UN countries) • Nationality (UN countries) • ID Issuing Country (UN countries) • Annual Income (Intervals: 0-10k, 10k-50k, 50k-100k, 100k+) • Profession/Activity (Standard, Cash-intensive, High-risk) • Marital Status (Single, Married, Divorced, Widowed, Separated)	• For each dropdown, only the pre-defined values will be selectable. No free typing allowed	**Must-Have**
FR-03	Intake Interface	REQ-01	**Data Validation & Field Formatting Rules:** • Client Name: Text string, max 100 characters • Client ID Number: Text String, max 100 characters • Officer Internal ID: Exactly 10-digit integer string • Email Address: Standard validation • Phone Number: International dial prefix dropdown + numeric local number • Officer Comment: Free-text field, max 500 characters	• Given an Officer ID of 9 or 11 characters, system raises an inline error: 'Officer ID must be exactly 10 digits'. • Given an invalid email format (e.g. 'client@domain'), form prevents progression.	**Must-Have**
FR-04	Intake Interface	REQ-01	**Date of Birth Calendar Picker & Age Validation:** The UI provides a calendar picker for Date of Birth (DD-MM-YYYY). The system will reject any date in the future and any client who is not at least 18 years old.	• Given a date resulting in age < 18 or a future date, an error is triggered: 'Client must be at least 18 years of age'. • Given a valid past date resulting in age >= 18, input is accepted.	**Must-Have**
FR-05	Data Storing	REQ-02	**SQL Database:** When the onboarding has been approved, the system saves the client's profile in a SQL database with the following fields: internal_client_id (Primary Key), intake form attributes, officer_id, officer_comment, calculated_risk_score, assigned_tier, final_decision, and created_timestamp.	• Given a confirmed onboarding submission, querying 'SELECT * FROM clients WHERE client_id = ?' returns complete matching data. • Data saved only commits upon clicking the final decision button.	**Must-Have**

FR-06	Audit & Overview	REQ-02	**Client Overview & Audit Governance UI:** The application provides a simple table view displaying saved client profiles with a search function. Columns include: Internal Client ID, Client Name, Issuing Country, Risk Tier, Final Decision, Officer ID, and Submission Date.	• Given saved clients in the database, navigating to the overview page displays all records correctly populated in the table. • Records can be filtered or sorted by Client ID and Risk Tier.	**Must-Have**
FR-07	Scoring Engine	REQ-03	**Weighted Risk Scoring Engine:** The scoring engine computes a composite score using the weighted formula: Total Score = Sum(w_i * v_i) / Sum(w_i) where w_i is the configured field category weight (summing to 100%) and v_i in [1, 5] is the risk score of the selected parameter value. Final scores map to: • Low Risk: 1.00 <= Score < 2.50 • Medium Risk: 2.50 <= Score < 3.75	• Given a pre-configured reference test profile, engine output matches the calculated weighted average to 2 decimal places. • Given a calculated score of 2.49, system classifies tier as 'Low Risk'. • Given a calculated score of 2.50, system classifies tier as 'Medium Risk'.	**Must-Have**
FR-08	Intake Interface	REQ-04	**Real-Time Calculation (Field Change):** The UI recalculates and update the risk score in real-time on the onChange event of any input field without needing a page reload. Unselected fields are treated as zero weight in the calculation during live preview.	• Given any change in a scoring dropdown, the provisional risk score and tier update on screen within <= 500ms.	**Nice-to-Have**
FR-09	Scoring Engine	REQ-05	**Red-Flag Automatic Override:** If a hard red-flag is triggered (e.g., Sanction List match, PEP confirmed), the system immediately overrides the weighted mathematical calculation, force the overall risk tier to 'High Risk (Rejected)', and display a dedicated banner with the trigger reason.	• Given a profile with all lowest-risk inputs (v=1) but PEP or Sanction flag is checked, the overall score overrides to High Risk. • Banner displays text: 'AUTOMATIC HIGH RISK OVERRIDE: Sanctions / PEP Hit Detected'.	**Must-Have**
FR-10	Workflow	REQ-06	**Risk Tier Action Buttons:** The UI presents three action buttons ('Approve', 'Submit for Manual Review', 'Decline / Rejection') with states determined by the evaluated risk tier: • Low Risk: 'Approve' is enabled; 'Decline' and 'Manual Review' are disabled. • Medium Risk: 'Submit for Manual Review' is enabled; 'Approve' and 'Decline' are disabled. • High Risk / Red Flag: 'Decline / Rejection' is enabled; 'Approve'	• Given High Risk status, clicking 'Approve' is impossible (button greyed out / unclickable). • Given Medium Risk status, only 'Submit for Manual Review' submits the workflow transition.	**Must-Have**
FR-11	Data Accesing	Other / Intake	**Existing Client ID Check:** Before accepting a new client, the system checks if there is an existing database record for matching combination of ID Number and country issuing the ID. If a match is found, creation is blocked with an alert.	• Given an ID Number and Country matching an existing record, system halts save and alerts: 'A client record with this ID Number and Issuing Country already exists'.	**Must-Have**

FR-12	Intake Interface/Scoring Engine	REQ-02/03	**Display Results:** The complete result of the scoring engine will be displayed. Every value will be displayed together with its weight and score.	output matches the calculated weighted average to 2 decimal places. • Given a pre-configured reference test profile, the displayed result will always be the same	**Must-Have**	New
FR-13	Scoring Engine	REQ-03	**Changeable weight values:** The different weight values of each field will be changeable in a simple UI. The UI will have the name of the field and a dropdown with a value of 1-5. The new values will only be locked and saved after clicking on a "Save" button.	• The new weights are only applied after having saved them. Quitting the UI without saving, the old values will be kept. • Given a pre-configured reference test profile, the risk score will be different after having changed the weight values	**Nice-To-Have**	New
FR-14	Scoring Engine	REQ-03	**Changeable Risk values:** The different risk values of each selecteable choice inside a field will be changeable in a simple UI. The UI will have the name of the field and each selecteable choice and their risk value. The new values will only be locked and saved after clicking on a "Save" button.	• The new risk values are only applied after having saved them. Quitting the UI without saving, the old values will be kept. • Given a pre-configured reference test profile, the risk score will be different after having changed the risk values	**Nice-To-Have**	New
FR-15	Audit	REQ-02	**Keeping Logs:** If an employee changes the weights of the fields, their internal ID, a new version number, and the changes will be documented in a table in the database. After having changed and saved the values, the risk scoring engine will use the latest version from the table to calculate the final score.	• Once the weights have been changed, the time of the change and the employee's ID have to be traceable in the logs file. • Each finalised onboarding connects to a version of the weights	**Nice-to-Have**	New